Suriyanarayanan V G

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.18
Kendriya Vidhyalaya Pattom	Class XII (CBSE)	2020	92.80%
Kendriya Vidhyalaya Pattom	Class X (CBSE)	2018	93.40%

RESEARCH EXPERIENCE

Development of Transient Absorption Spectrometer | IIT Hyderabad

- Instrumented a **femtosecond transient absorption** setup for ultrafast optical measurements, integrating spectrometer, detector systems, and optical components to establish a functional platform for time-resolved spectroscopy.
- Automated **translation-stage control** for a femtosecond transient absorption spectrometer using **Thorlabs Kinesis .NET and LabVIEW**, implementing programmable scan ranges, equidistant positioning, and automated stage movement, with synchronized transient absorption data acquisition and position-linked data logging.
- Developed **LabVIEW VIs** for instrument initialization, memory allocation, camera configuration, hardware interfacing, integrated detector systems and acquisition hardware to support synchronized operation and data acquisition.
- Integrated an **optical chopper** and **photodetector** into the transient absorption setup to enable periodic pump-beam modulation and detection of pump-induced probe intensity changes for differential absorption measurements.

Fabrication of CsPbBr Perovskite Photodetector Device | IIT Hyderabad

- Synthesized **CsPbBr perovskite nanocrystals using the hot-injection method**, establishing the material synthesis process for subsequent characterization and integration into optoelectronic devices.
- Fabricated a **solution-processable CsPbBr perovskite photodetector** using an FTO/TiO/CsPbBr device architecture, integrating the transparent conducting substrate, electron transport layer, and photoactive perovskite layer.
- Employed **spin coating for CsPbBr thin-film deposition** and **blade coating for carbon-paste electrode formation**, utilizing solution-based processing techniques to assemble the photodetector device.
- Performed **electrical characterization** of CsPbBr perovskite photodetector using a Keithley source meter, measuring device electrical response to evaluate its operating characteristics and suitability for optoelectronic applications.

INTERNSHIPS

Multi-Channel EMG Signal Acquisition System | NIT Tiruchirappalli

- Engineered an **8-channel EMG acquisition system** incorporating instrumentation amplifiers and low-noise analog front-end circuitry for multichannel biopotential signal acquisition, conditioning, and reliable recording.
- Designed and developed the **EMG instrumentation chain in KiCad**, integrating INA333 instrumentation amplifiers, OPA2333-based filtering stages, analog multiplexing, and level-shifting circuits for multichannel signal conditioning.
- Integrated **ESP32, 16-bit ADC, and analog multiplexing circuitry** for multichannel EMG data acquisition and real-time processing, incorporating signal conditioning, grounding, and noise-reduction techniques.

Small Satellite Launch Vehicle Control Actuation System

- Executed **SS2/SS3 SSLV control-system testing**, validating pitch and yaw responses through the flex-nozzle system.
- Applied the **ADSP2106** for control-system implementation and signal processing in the launch vehicle actuation system.
- Tested **ADC/DAC signal paths across prime and redundant control circuits** to verify reliable signal transmission and control-system operation, while coordinating with the **Avionics Entity (AVN) at VSSC** during SSLV control-actuation testing and system-level performance evaluation.

PROJECTS

Model Predictive Control for PMSM Speed Control

- Achieved an **88.02% improvement in steady-state error** by benchmarking Model Predictive Control (MPC) against conventional PI control for PMSM speed regulation under comparable simulation conditions.
- Integrated **Model Predictive Control (MPC) with Field-Oriented Control (FOC)** to regulate Permanent Magnet Synchronous Motor (PMSM) speed, enabling controlled motor operation in response to changing operating conditions.
- Developed and simulated an **MPC-based PMSM speed-control model in MATLAB/Simulink**, then implemented the control algorithm on a **TI TMS320F28069 DSP** for embedded motor-control applications.
- Validated **sensorless Field-Oriented Control using a Sliding Mode Observer (SMO)** to estimate rotor position and speed, enabling feedback-based motor operation with a switching threshold of approximately **170 RPM**.

Quantum Computation with Quantum Dots

- Analyzed the **Loss-DiVincenzo quantum-dot qubit proposal**, exploring quantum confinement, electron tunneling, and exchange interactions for implementing **SWAP and SWAP gates** in spin-based quantum computation.
- Modeled **two-electron exchange coupling** using the Heisenberg Hamiltonian, deriving singlet and triplet spin states and analyzing their time evolution to understand exchange-driven dynamics in quantum-dot qubits.
- Implemented a **two-qubit SWAP gate in Qiskit**, simulating quantum-state evolution and analyzing exchange-mediated state transfer between qubits, with gate behavior evaluated through **1024-shot measurements**.

Chassis Balancing using PID Controllers

- Engineered a **self-balancing robot** using an ESP32 and MPU6050 IMU, implementing complementary filtering to achieve approximately **0.05° stationary tilt-estimation noise** for stable orientation measurement.
- Implemented PID control by tuning of Kp, Ki, and Kd to suppress oscillations and achieve smooth closed-loop balancing.
- Diagnosed **H-bridge motor driver failures** during robot testing and upgraded the motors from **60 RPM to 300 RPM**, improving motor responsiveness and supporting more effective closed-loop balancing control.
- Validated **real-time robot balancing** using serial telemetry and PWM-based motor control within a **1,550 hardware budget**.

SEMINARS

Plasmonic Quantum Dot Based Devices

Future With India Program (FWIP) Selection Process

Superconducting MEMS Resonator

Course Seminar, Indian Institute of Technology Hyderabad

Quantum Cascade Laser

Course Seminar, Indian Institute of Technology Hyderabad

Model Predictive Speed Control of PMSM Motor

B.Tech Final Year Seminar

Robust Speed Control of PMSMUndergraduate Technical Seminar

Sree Chitra Thirunal College of Engineering

Quantum Sensing Devices

Course Seminar, Indian Institute of Technology Hyderabad

AREA OF EXPERTISE

Programming & Tools: Python, MATLAB/Simulink

Instrumentation & Measurement: LabVIEW, Optical Instrumentation, Transient Absorption Spectroscopy, EMG Signal Acquisition

Electronics & Signal Processing: Analog Signal Conditioning, Instrumentation Amplifiers, ADCs, Analog Filtering, Noise Reduction, Sensor Interfacing

Control & Modeling: Model Predictive Control (MPC), Proportional-Integral-Derivative Control (PID), Field-Oriented Control (FOC), Mathematical Modeling and Simulation

POSITION OF RESPONSIBILITY

Teaching Assistant for Course EP1108, IIT Hyderabad

Team Leader, TATA Elxsi TELEPORT Competition- Led the college-finalist team.

Project Team Leader & ISRO Point of Contact, B.Tech Final Year Project- Coordinated communication between the project team, college, and ISRO.

Event Organizer, Optica, IIT Hyderabad - Organized and conducted an Optica outreach event at DAV School, IIT Hyderabad.

ACHIEVEMENTS

Selected for the **Future with India Program**, a Japan-based research exchange and internship program.

College Finalist, Tata Elxsi TELEPORT - National-level idea challenge for engineering students.

Gold & Silver Medallist - Milan and DIESTA Powerlifting Competition.

EXTRACURRICULAR ACTIVITIES & INTERESTS

Peer Fitness Mentor: Guided 12 peers through personalized strength-training and weight-management routines.

Anabolic Cooking: Reworked high-calorie meals into customized low-calorie, high-protein recipes.

Powerlifting: Strength training and powerlifting.

Creative Design: Customized T-shirts and phone cases using hand-drawn artwork.

Video Editing: Create and edit short-form video content.